Deriving Normal Equation for Multiple Linear Regression

Bhanumathi Ramesh · Follow

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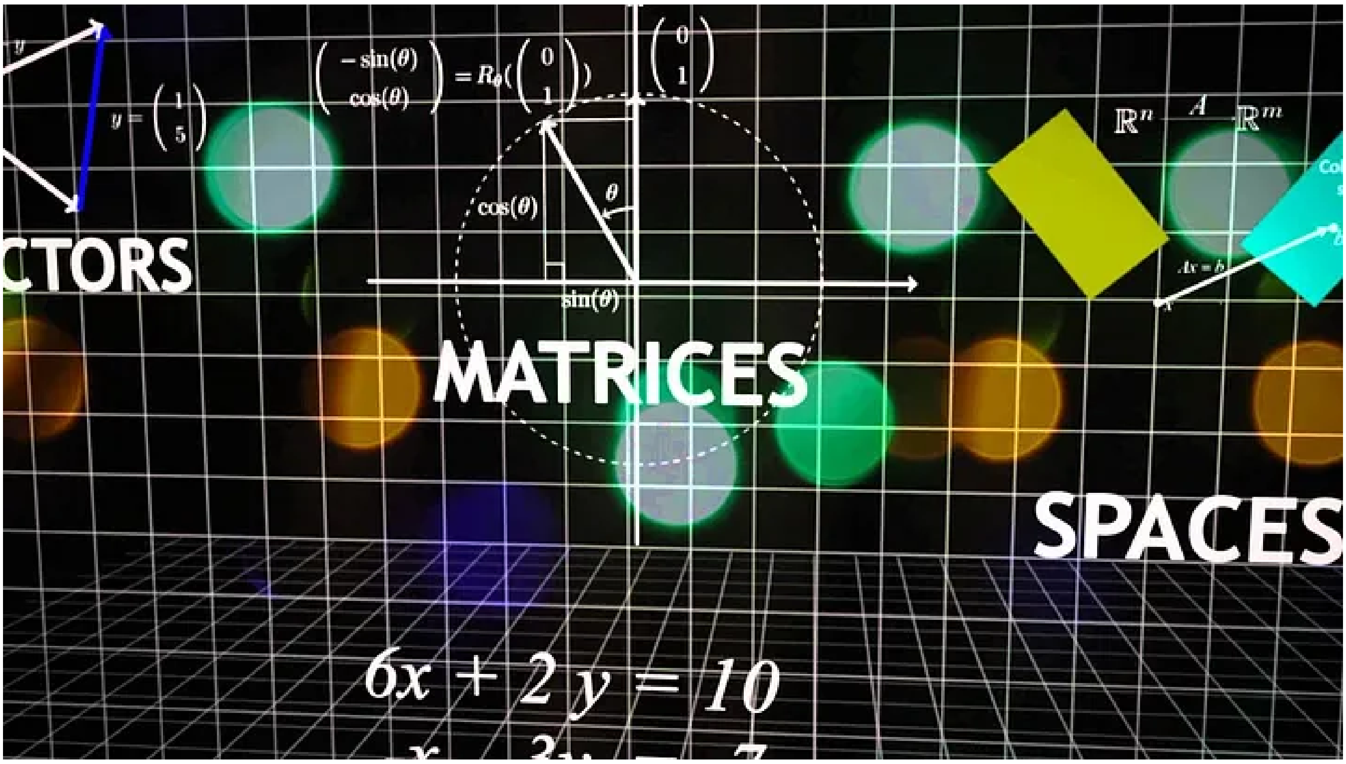
The mathematical formulation of Normal Equation



In this article, we will be seeing the steps to derive the normal equation for multiple linear regression.

What is Multiple Linear Regression?

Multiple Linear Regression is a supervised machine learning algorithm. This algorithm uses more than one independent variable to predict a target variable. The below table is an example of a multiple linear regression problem.

	General_knowledge	Aptitude	Mathematics	Science	SAT
0	73	71	74	73	144
1	93	90	60	97	186
2	89	94	97	98	182
3	96	93	115	110	208
4	73	68	87	83	157

independent variable(as called predictors,explanatory variable or regressor (in regression problem)) and one dependent variable (also called label,target,response variable). Since we have more than one independent variable we have to find the **coefficients** for each independent variable. In my Simple Linear Regression article, we have discussed how to find the coefficients. Now we will extent simple linear to multiple linear and find the coefficients.

Mathematical Derivation

We know the equation of a line for simple linear regression is

$$\hat{y} = \beta_1 x + \beta_0$$

Equation of a line

Similarly, now for the multiple linear regression, the formula will be,

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 \text{-----} \beta_n x_n$$

Equation of a plane with multiple independent variables

The above formula can be rewritten in matrix form as,

$$Y = \begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \hat{y}_3 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} \beta_0 & \beta_1 x_{11} & \beta_2 x_{12} & \beta_3 x_{13} & \cdots & \beta_m x_{1m} \\ \beta_0 & \beta_1 x_{21} & \beta_2 x_{22} & \beta_3 x_{23} & \cdots & \beta_m x_{2m} \\ \beta_0 & \beta_1 x_{31} & \beta_2 x_{32} & \beta_3 x_{33} & \cdots & \beta_m x_{3m} \\ \vdots & & & & & \\ \beta_0 & \beta_1 x_{n1} & \beta_2 x_{n2} & \beta_3 x_{n3} & \cdots & \beta_m x_{nm} \end{bmatrix}$$

Matrix representation for n records

This can be further rewritten as,

$$\begin{bmatrix} \hat{Y}_1 \\ \hat{Y}_2 \\ \hat{Y}_3 \\ \vdots \\ \hat{Y}_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & x_{12} & x_{13} & \cdots & x_{1m} \\ 1 & x_{21} & x_{22} & x_{23} & \cdots & x_{2m} \\ 1 & x_{31} & x_{32} & x_{33} & \cdots & x_{3m} \\ \vdots & & & & & \\ 1 & x_{n1} & x_{n2} & x_{n3} & \cdots & x_{nm} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \\ \vdots \\ \beta_m \end{bmatrix}$$

Now the above matrix can be represented as (all capital letters represents Matix in further equations)

$$Y = X\beta$$

matrix representation

we know, the error is given by (Actual — Predicted), So the error can be represented in matrix form as below.

$$e = Y - \hat{Y}$$

$$Y = \begin{bmatrix} Y_1 \\ Y_2 \\ Y_3 \\ \vdots \\ Y_n \end{bmatrix} \quad e = \begin{bmatrix} Y_1 - \hat{Y}_1 \\ Y_2 - \hat{Y}_2 \\ Y_3 - \hat{Y}_3 \\ \vdots \\ Y_n - \hat{Y}_n \end{bmatrix}$$

we know, for linear regression, for n observations, the **error/cost function** is given by,

$$E = \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 \quad \text{—— (1)}$$

Error/cost function

Lets, derive an equation for the error function to represent in matrix form

Expanding the equation 1, we can rewrite it as,

$$E = (Y_1 - \hat{y}_1)^2 + (Y_2 - Y_2)^2 + (Y_3 - \hat{Y}_3)^2 \cdots\cdots (Y_n - \hat{Y}_n)^2$$

Expanding the error function

The above equation can be rewritten in matrix form as,

$$\begin{bmatrix}(y_1-y_1) & (y_2-y_2) & (y_3-y_3)^2 & \cdots & (y_n-y_n)\end{bmatrix}\begin{bmatrix}(y_1-\hat{y}_1)\\(y_2-\hat{y}_2)\\(y_3-\hat{y}_3)\\\vdots\\(y_n-\hat{y}_n)\end{bmatrix}$$

Representing error function in matrix form

Hence, the Error/Cost function in matrix notation is given by,

$$E = e^T e$$

Error Function in Matrix notation

Now the error equation becomes,

$$E = (Y-\hat{Y})^T (Y-\hat{Y}) \quad \text{---} \; (2).$$

By Linear algebra,

$$(A+B)^T = A^T + B^T$$
$$(A-B)^T = A^T - B^T$$

Substituting the values in equation 2, equation 2 can be rewritten as,

$$E = (Y^T - \hat{Y}^T)(Y-\hat{Y}) \quad \text{---} \; (3).$$

Replacing, Yhat = Xβ in equation 3,

$$E = \left(Y^T - (X\beta)^T\right)\left(Y - X\beta\right)$$

$$= Y^TY - Y^TX\beta - Y(X\beta)^T + (X\beta)^TX\beta \quad \text{—④}$$

To simplify equation 4, we have to prove the following term as equal

$$= (X\beta)^TY = Y^TX\beta \quad \text{—⑤}$$

solving equation 5,

Proving above Equation ⑤.

Let $Y = A$

$X\beta = B$

Equation ⑤ will be

$$A^T\beta = B^TA \quad \text{—⑥}$$

By linear algebra

$$(AB)^T = B^TA^T, \quad (A+B)^T = A^T + B^T$$

$$(A^TB)^T = B^TA, \quad (A-B)^T = A^T - B^T$$

Now Equation 6 will be

$$A^TB = (A^TB)^T$$

$$Y^TX\beta = (Y^TX\beta)^T$$

after simplifying, equation 4 becomes,

$$E = Y^TY - 2Y^TX\beta + \beta^TX^TX\beta \quad \text{—⑦}$$

Applying partial derivatives concerning β,

$$\frac{d(\epsilon)}{d\beta} = \frac{d}{d\beta}\left(Y^TY - 2Y^TX\beta + \beta^TX^TX\beta\right)$$
$$= 0 - 2Y^TX + \frac{d}{d\beta}\left(\beta^TX^TX\beta\right) \quad \text{— (8)}$$

By applying matrix differentiation on equation 8,

$$-2Y^TX + 2X^TX\beta^T = 0$$
$$X^TX\beta^T = Y^TX$$

simplifying the equation further,

$$\beta^T = \frac{Y^TX}{X^TX}$$
$$\beta^T = Y^TX\left(X^TX\right)^{-1}$$
$$\beta = \left[Y^TX\left(X^TX\right)^{-1}\right]^T$$
$$\beta = \left(\left(X^TX\right)^{-1}\right)^T\left(Y^TX\right)^T$$
$$\beta = \left(X^TX^{-1}\right)\left(X^TY\right)$$

After simplifying we have got a β coefficient matrix.

$$\beta = \left(X^TX^{-1}\right)\left(X^TY\right)$$

In the above equation X represents X_train, Y represents y_train from the dataset. Plugging those values into the above equations will give us the β coefficient matrix. This matrix will have (independent variables + 1) different values. Suppose if your data set has 5 independent variables, your coefficients will be 6.

Predictions can be made using X_Test and β coefficient. That will be the dot product of X_test and β coefficient.

Conclusion

This normal equation is used in Sci-kit learn's LinearRegression() which works well for small datasets. LinearRegression()'s computational complexity is too high since it involves calculating matrix inverse operation and complexity is $o(n^3)$. Please refer complexity of matrix inverse operation. To overcome these complexity issues, SGDRegressor() method is used, which uses gradient descent to find the coefficients.

Please refer to the practical implementation of Normal Equation here — **https://git.io/J6Wes**

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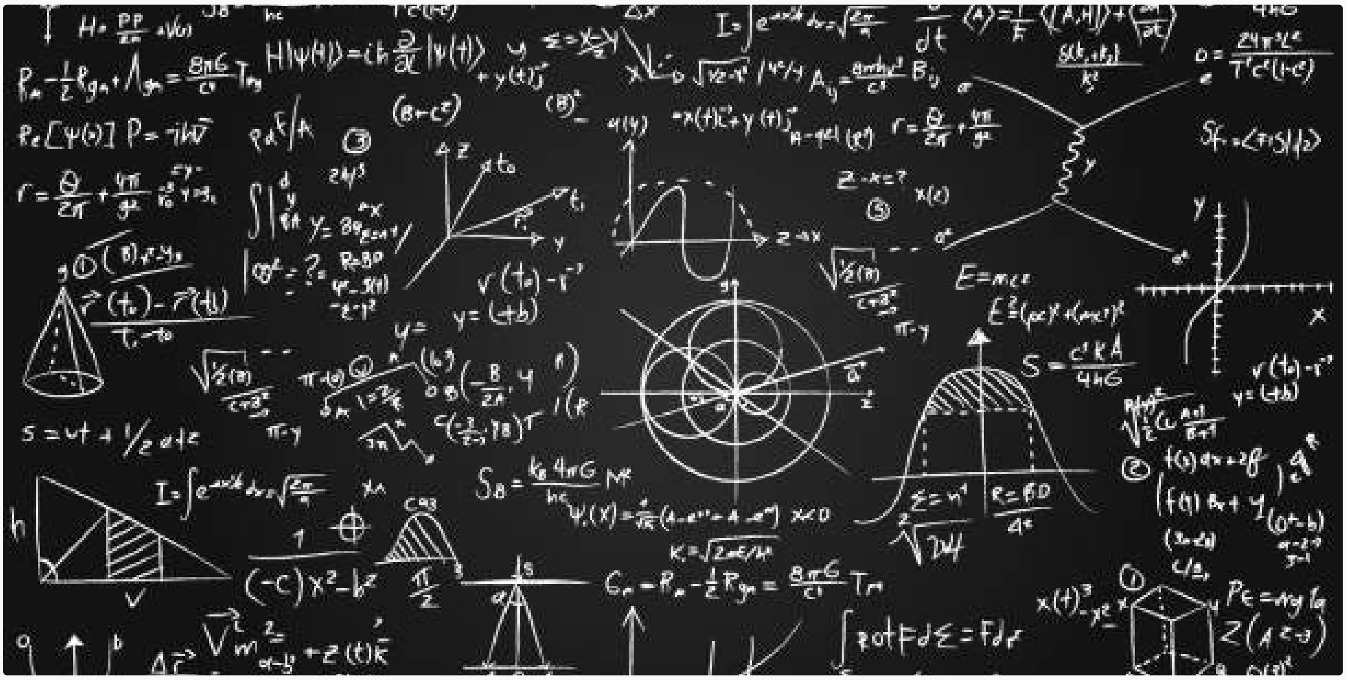

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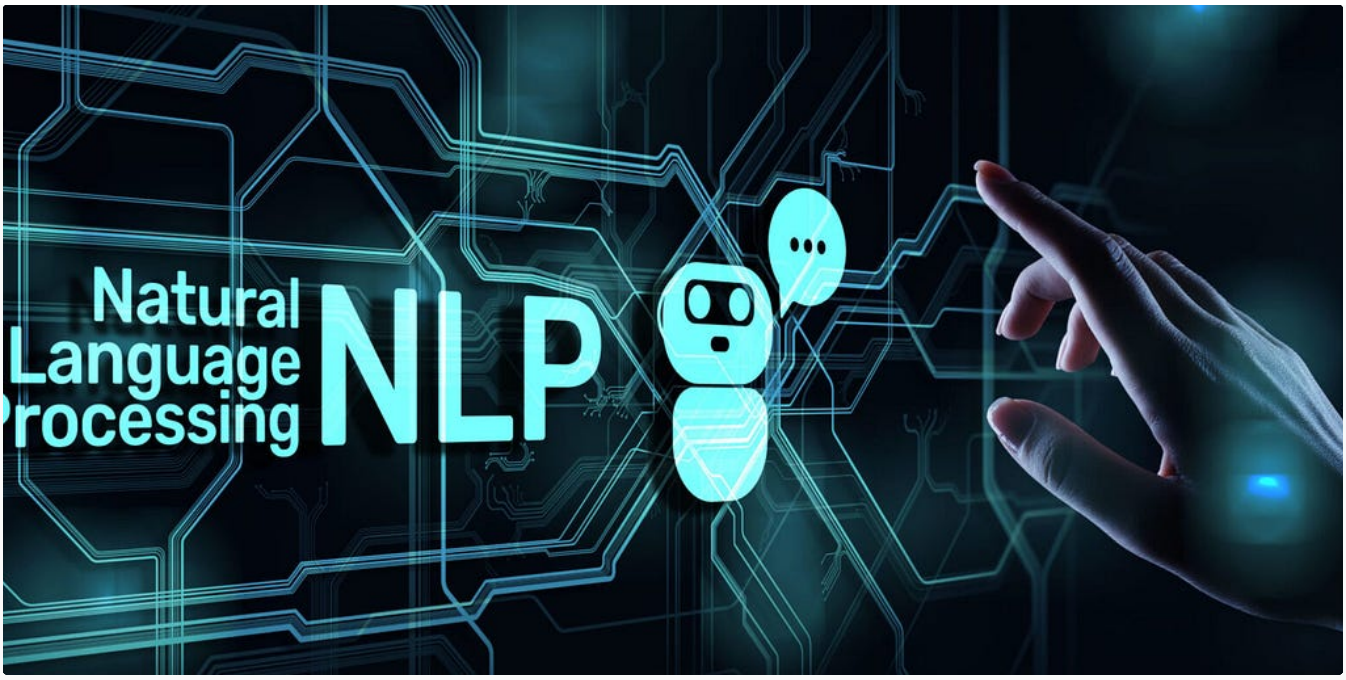

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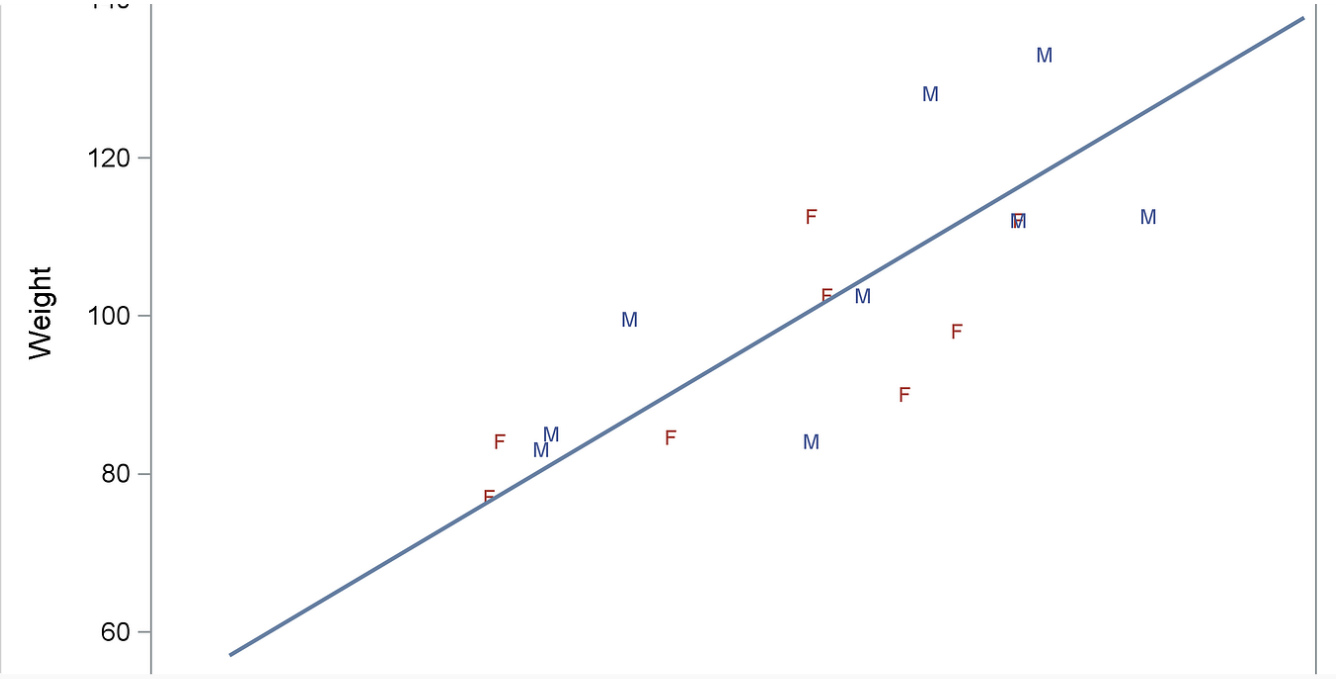

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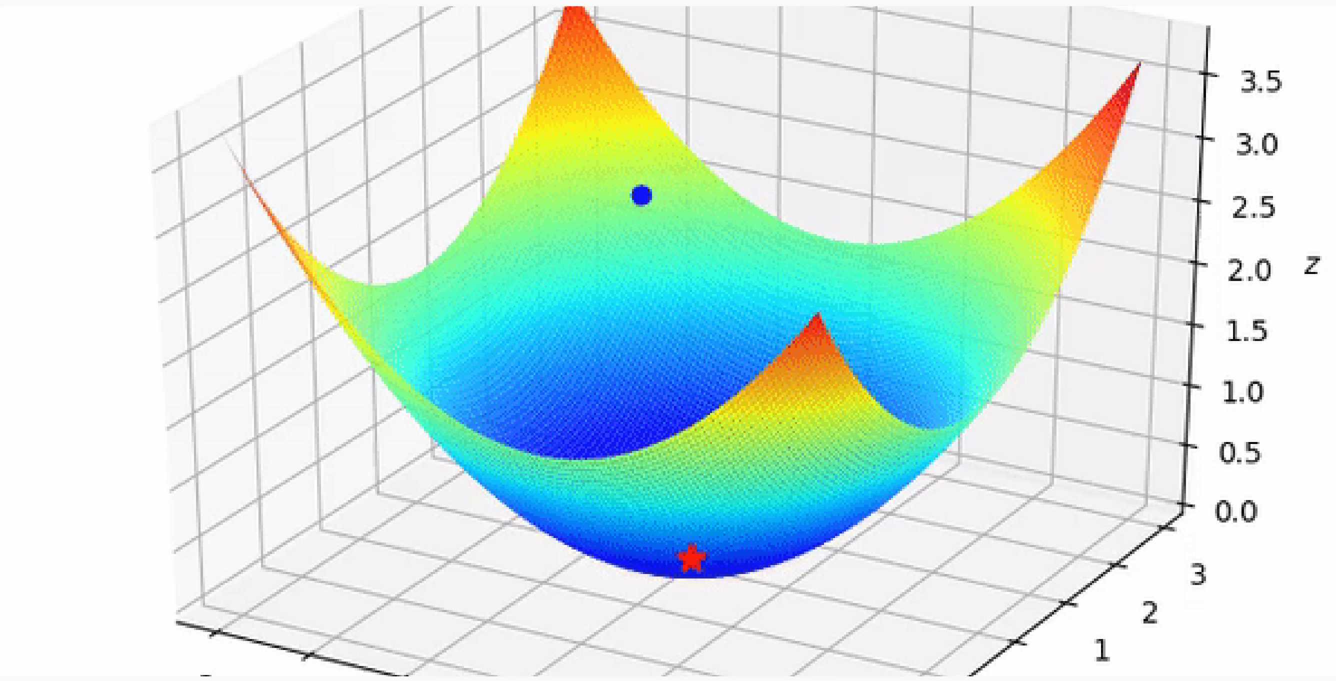

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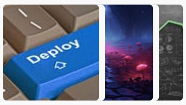

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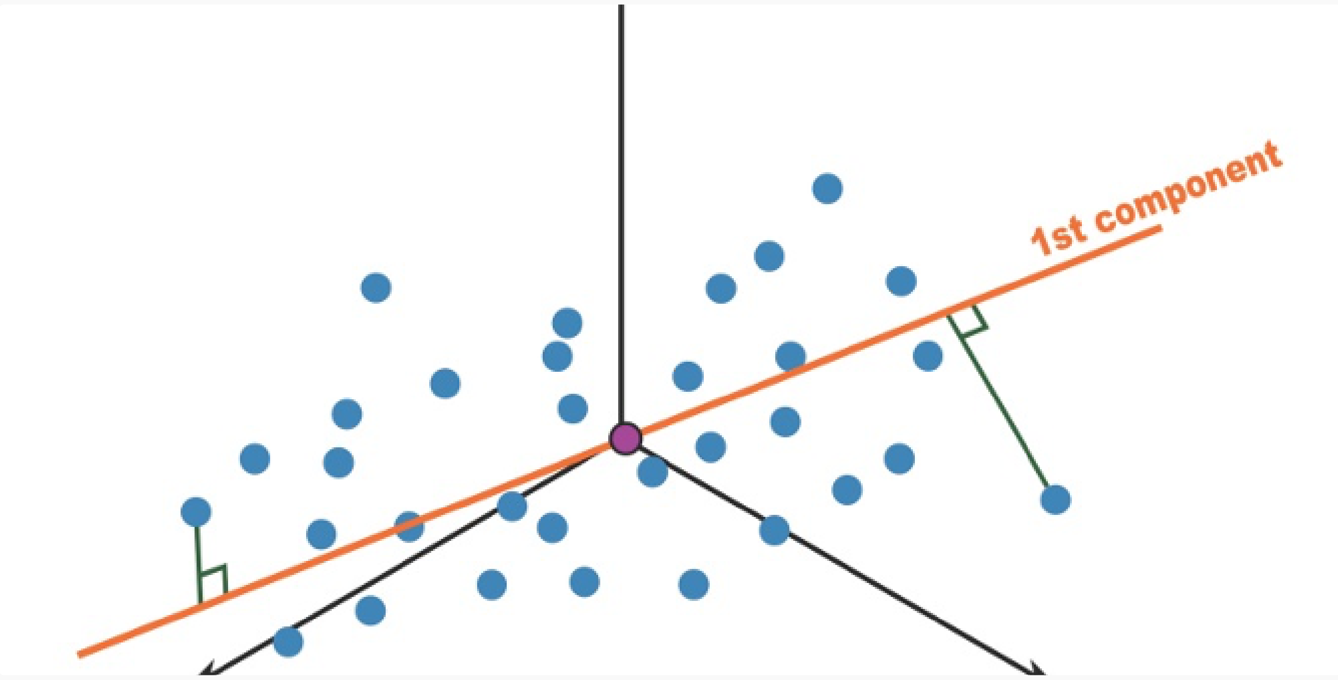

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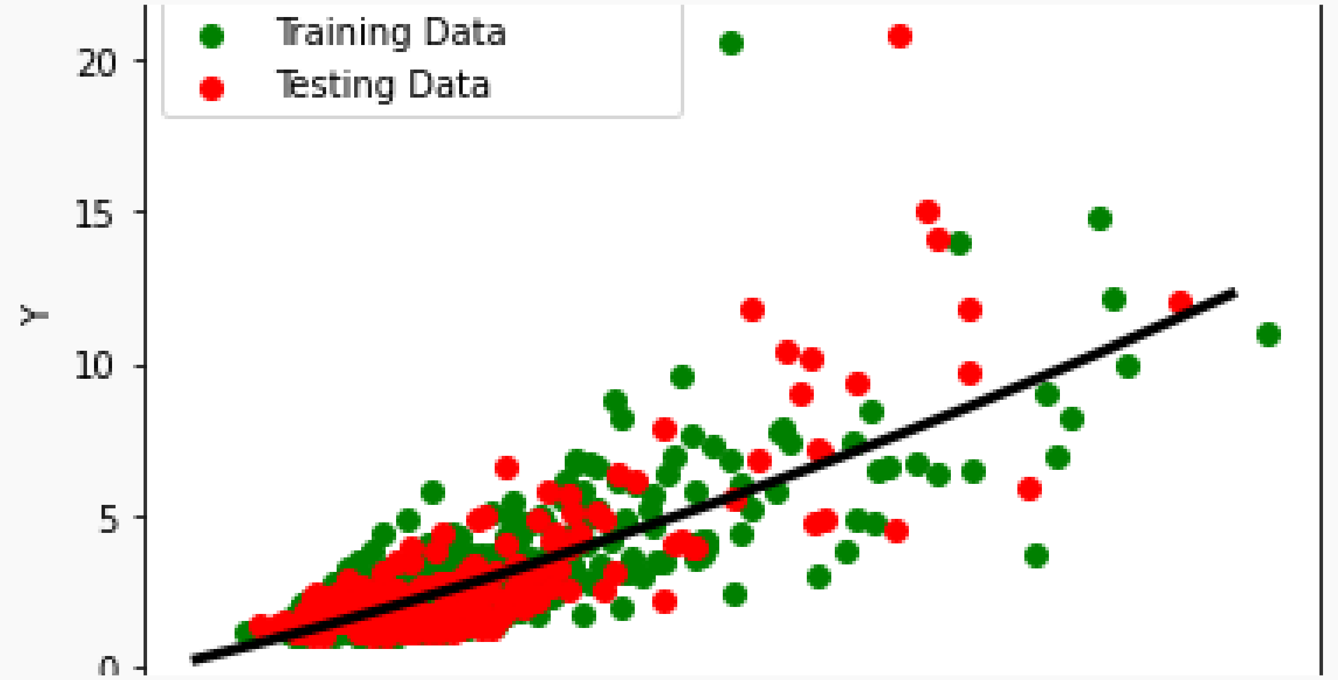

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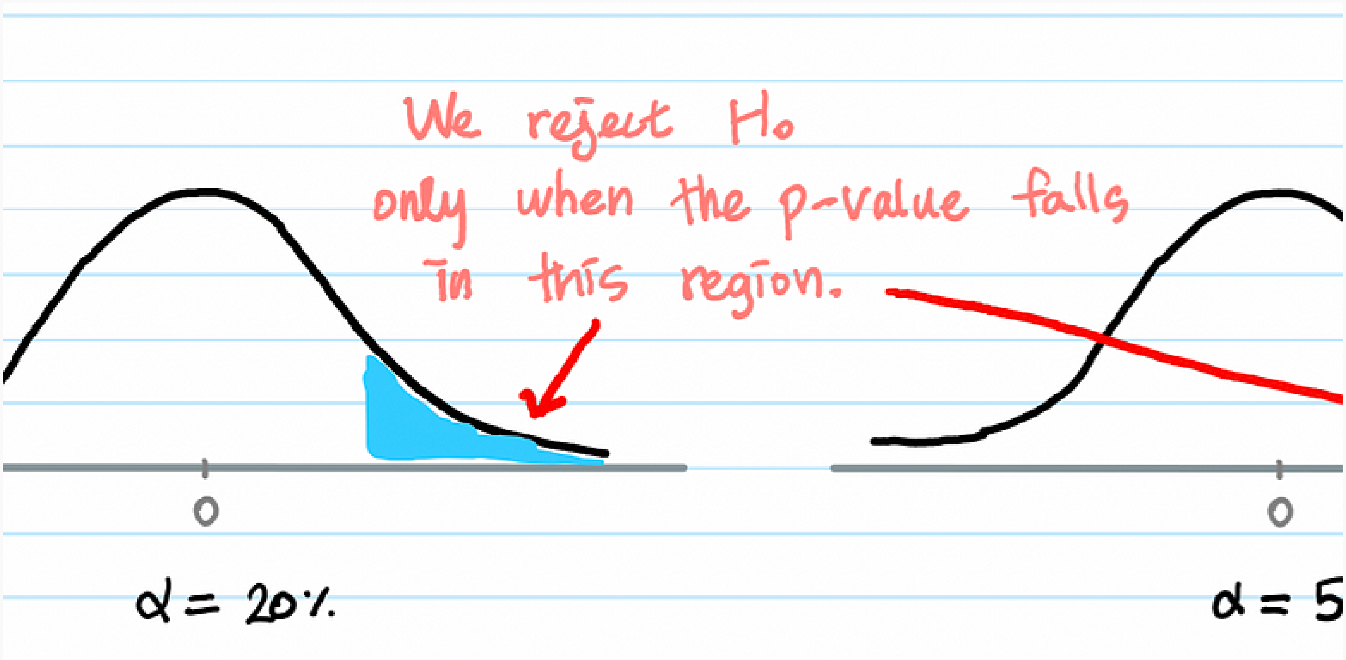

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